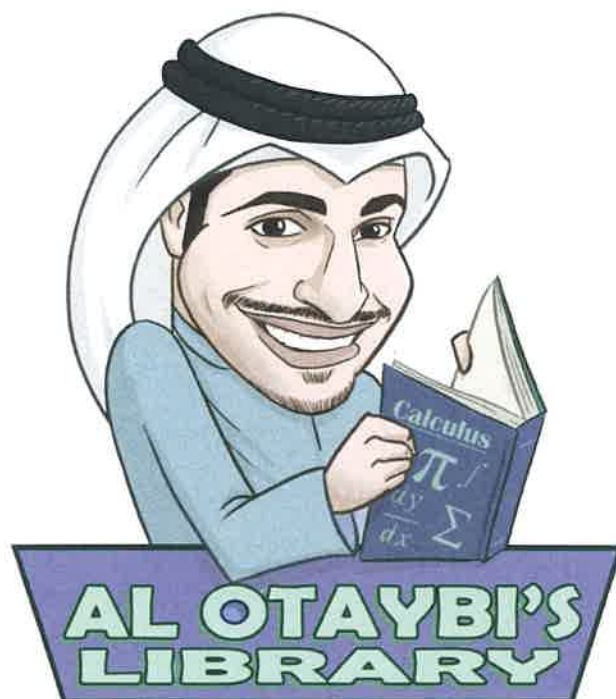


Kuwait University

2021/2022

Fall course



Calculus A

First

Section : 2.3 (part a)

Calculating Limits Using the Limit Laws

(5)



Special case

عند وجود المطلق



التعويض المباشر



خطوات حل مسائل اللمت

$\sin(\infty)$
 $\cos(\infty)$

$\dots \leq f(x) \leq \dots$

Sandwich
Theorem

∞
 $—$
 ∞

تحليل خاص !!

∞

نبتل اللمت
بالاتجاهين

$\frac{0}{0}$, $0 \cdot \infty$

التحليل

مربعين-مكعبين-عامل مشترك
تحليل حدودية من الدرجة الثانية
توحيد مقامات — ملحق —> حل

$\infty — \infty$

الضرب بالمرافق

توحيد مقامات

شكل المرافق

$\sqrt{\quad} — \sqrt{\quad}$
عدد — عدد
عدد — عدد

رقعة

Stop !!

Note

$\frac{0}{\text{عدد}} = 0$
 $\frac{\text{عدد}}{0} = \infty$
 $\frac{\text{عدد}}{\text{عدد}} = 0$



Evaluate the limit if it exist



$$\lim_{x \rightarrow 2} \frac{x^2 - 1}{x + 3}$$

$$= \frac{(2)^2 - 1}{2 + 3} = \frac{4 - 1}{5} = \frac{3}{5}$$

$$\lim_{x \rightarrow 0} \frac{\sin^{-1} x - 3}{e^x}$$

$$= \frac{\sin^{-1} 0 - 3}{e^0} = \frac{0 - 3}{1} = \frac{-3}{1} = -3$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x + 1}$$

$$= \frac{\cos 0 - 1}{0 + 1} = \frac{1 - 1}{1} = \frac{0}{1} = 0$$

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$$

$$\frac{4 - 4}{2 - 2} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x+2)}{\cancel{x-2}}$$

$$= \lim_{x \rightarrow 2} (x+2) = 2+2 = 4$$

$$\lim_{x \rightarrow -2} \frac{x+2}{x^3 + 8}$$

$$= \frac{-2+2}{-8-8} = \frac{0}{-16}$$

$$= \lim_{x \rightarrow -2} \frac{\cancel{x+2}}{\cancel{(x+2)}(x^2 - 2x + 4)}$$

$$= \lim_{x \rightarrow -2} \frac{1}{x^2 - 2x + 4} = \frac{1}{4 + 4 + 4} = \frac{1}{12}$$

$$\lim_{x \rightarrow 0} \frac{x^2 + 2x}{x}$$

$$\frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{x}(x+2)}{\cancel{x}}$$

$$= \lim_{x \rightarrow 0} (x+2) = 0+2 = 2$$

$$\lim_{x \rightarrow -1} \frac{x+1}{x^2 - 2x - 3}$$

$$\frac{-1+1}{1+2-3} = \frac{0}{0}$$

$$= \lim_{x \rightarrow -1} \frac{\cancel{x+1}}{(\cancel{x+1})(x-3)}$$

$$= \lim_{x \rightarrow -1} \frac{1}{x-3} = \frac{1}{-1-3} = \frac{1}{-4} = -\frac{1}{4}$$

$$\lim_{x \rightarrow 0} \frac{4 - \sqrt{16+x}}{x}$$

$$\frac{4 - \sqrt{16+0}}{0} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{4 - \sqrt{16+x}}{x} \cdot \frac{4 + \sqrt{16+x}}{4 + \sqrt{16+x}}$$

$$= \lim_{x \rightarrow 0} \frac{16 - [16+x]}{x [4 + \sqrt{16+x}]}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{16} - \cancel{16} - x}{x [4 + \sqrt{16+x}]}$$

$$= \lim_{x \rightarrow 0} \frac{-x}{\cancel{x} [4 + \sqrt{16+x}]}$$

$$= \lim_{x \rightarrow 0} \frac{-1}{4 + \sqrt{16+x}} = \frac{-1}{4 + \sqrt{16}} = \frac{-1}{4+4} = -\frac{1}{8}$$

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 3x + 2}$$

0/0

$$= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(x^2 + x + 1)}{\cancel{(x-1)}(x-2)}$$

$$= \frac{3}{-1} = -3$$

$$\lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x^3 - 8}$$

0/0

$$= \lim_{x \rightarrow 2} \frac{\sqrt{x} - \sqrt{2}}{x^3 - 8} * \frac{\sqrt{x} + \sqrt{2}}{\sqrt{x} + \sqrt{2}}$$

$$= \lim_{x \rightarrow 2} \frac{x - 2}{(x^3 - 8) [\sqrt{x} + \sqrt{2}]}$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{x-2}}{\cancel{(x-2)}(x^2 + 2x + 4) [\sqrt{x} + \sqrt{2}]}$$

$$= \frac{1}{(4+4+4) [\sqrt{2} + \sqrt{2}]} = \frac{1}{12 (2\sqrt{2})} = \frac{1}{24\sqrt{2}}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{2x^3 - 1} - 1}{x - 1}$$

$$\frac{\sqrt{1} - 1}{1 - 1} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\sqrt{2x^3 - 1} - 1}{x - 1} * \frac{\sqrt{2x^3 - 1} + 1}{\sqrt{2x^3 - 1} + 1}$$

$$= \lim_{x \rightarrow 1} \frac{2x^3 - 1 - 1}{(x - 1) [\sqrt{2x^3 - 1} + 1]}$$

$$= \lim_{x \rightarrow 1} \frac{2x^3 - 2}{(x - 1) [\sqrt{2x^3 - 1} + 1]}$$

$$= \lim_{x \rightarrow 1} \frac{2(x^3 - 1)}{(x - 1) [\sqrt{2x^3 - 1} + 1]}$$

$$= \lim_{x \rightarrow 1} \frac{2(x - 1)(x^2 + x + 1)}{(x - 1) [\sqrt{2x^3 - 1} + 1]}$$

$$= \frac{2(1 + 1 + 1)}{\sqrt{2 - 1} + 1} = \frac{2(3)}{\sqrt{1} + 1} = \frac{6}{2} = 3$$

$$3) \lim_{x \rightarrow 1} \frac{\sqrt{x+8} - 3}{x^2 + 8x - 9} = \frac{\sqrt{9} - 3}{1 + 8 - 9} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{x+8} - 3}{x^2 + 8x - 9} * \frac{\sqrt{x+8} + 3}{\sqrt{x+8} + 3}$$

$$\lim_{x \rightarrow 1} \frac{x + 8 - 9}{(x^2 + 8x - 9)(\sqrt{x+8} + 3)}$$

$$\lim_{x \rightarrow 1} \frac{x - 1}{(x^2 + 8x - 9)(\sqrt{x+8} + 3)}$$

$$\lim_{x \rightarrow 1} \frac{\cancel{x - 1}}{(\cancel{x - 1})(x + 9)(\sqrt{x+8} + 3)}$$

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الحد الصفري

$$\lim_{x \rightarrow 1} \frac{1}{(x + 9)(\sqrt{x+8} + 3)}$$

$$= \frac{1}{(10)(\sqrt{9} + 3)}$$

$$= \frac{1}{10 \cdot (3 + 3)} = \frac{1}{10(6)} = \frac{1}{60}$$

$$4) \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x^2 - 1}$$

$$= \frac{1-1}{1-1} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x^2 - 1} * \frac{\sqrt{x} + 1}{\sqrt{x} + 1}$$

$$\lim_{x \rightarrow 1} \frac{x - 1}{(x^2 - 1)(\sqrt{x} + 1)}$$

$$\lim_{x \rightarrow 1} \frac{\cancel{x - 1}}{(\cancel{x - 1})(x + 1)(\sqrt{x} + 1)}$$

$$\lim_{x \rightarrow 1} \frac{1}{(x + 1)(\sqrt{x} + 1)}$$

$$= \frac{1}{(1 + 1)(\sqrt{1} + 1)}$$

$$= \frac{1}{2(2)} = \frac{1}{4}$$

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الحد الحرفي

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Text book questions



$$\lim_{t \rightarrow -2} \frac{t^4 - 2}{2t^2 - 3t + 2}$$

$$= \frac{16 - 2}{2(4) + 6 + 2}$$

$$= \frac{14}{8 + 8}$$

$$= \frac{14}{16} = \frac{7}{8}$$

$$\lim_{t \rightarrow 2} \left(\frac{t^2 - 2}{t^3 - 3t + 5} \right)^2$$

$$= \left(\frac{4 - 2}{8 - 6 + 5} \right)^2$$

$$= \left(\frac{2}{7} \right)^2$$

$$= \frac{4}{49}$$

$$\lim_{x \rightarrow 3} \frac{x - 3}{x^3 - 27} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 3} \frac{\cancel{x-3}}{(\cancel{x-3})(x^2 + 3x + 9)}$$

$$= \frac{1}{9 + 9 + 9}$$

$$= \frac{1}{27}$$

$$\lim_{t \rightarrow 1} \frac{t^4 - 1}{t^3 - 1} \quad \frac{0}{0}$$

$$\lim_{t \rightarrow 1} \frac{(t^2 - 1)(t^2 + 1)}{(t - 1)(t^2 + t + 1)}$$

$$\lim_{t \rightarrow 1} \frac{\cancel{(t-1)}(t+1)(t^2+1)}{\cancel{(t-1)}(t^2+t+1)}$$

$$= \frac{(2)(2)}{3}$$

$$= \frac{4}{3}$$



Text book questions



$$\lim_{t \rightarrow 0} \left(\frac{\sqrt{1+t} - \sqrt{1-t}}{t} \right)$$

$\frac{0}{0}$

$$= \lim_{t \rightarrow 0} \frac{\sqrt{1+t} - \sqrt{1-t}}{t} \times \frac{\sqrt{1+t} + \sqrt{1-t}}{\sqrt{1+t} + \sqrt{1-t}}$$

$$= \lim_{t \rightarrow 0} \frac{1+t - (1-t)}{t [\sqrt{1+t} + \sqrt{1-t}]} = \lim_{t \rightarrow 0} \frac{1+t - 1 + t}{t [\sqrt{1+t} + \sqrt{1-t}]}$$

$$= \lim_{t \rightarrow 0} \frac{2t}{t [\sqrt{1+t} + \sqrt{1-t}]} = \frac{2}{1+1} = \frac{2}{2} = 1$$

$$\lim_{x \rightarrow 16} \left(\frac{4 - \sqrt{x}}{16x - x^2} \right)$$

$\frac{0}{0}$

$$= \lim_{x \rightarrow 16} \frac{4 - \sqrt{x}}{16x - x^2} \times \frac{4 + \sqrt{x}}{4 + \sqrt{x}}$$

$$= \lim_{x \rightarrow 16} \frac{16 - x}{(16x - x^2) [4 + \sqrt{x}]}$$

$$= \lim_{x \rightarrow 16} \frac{16 - x}{x (16 - x) [4 + \sqrt{x}]}$$

$$= \frac{1}{16 [4 + 4]} = \frac{1}{16 (8)} = \frac{1}{128}$$

$$\lim_{x \rightarrow 2} \frac{\sqrt{4-x} - \sqrt{x}}{x-2}$$

$$\frac{\sqrt{2} - \sqrt{2}}{2-2} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 2} \frac{\sqrt{4-x} - \sqrt{x}}{x-2} \cdot \frac{\sqrt{4-x} + \sqrt{x}}{\sqrt{4-x} + \sqrt{x}}$$

$$= \lim_{x \rightarrow 2} \frac{4-x-x}{(x-2) [\sqrt{4-x} + \sqrt{x}]}$$

$$= \lim_{x \rightarrow 2} \frac{4-2x}{(x-2) [\sqrt{4-x} + \sqrt{x}]}$$

$$= \lim_{x \rightarrow 2} \frac{-2(x-2)}{(x-2) [\sqrt{4-x} + \sqrt{x}]}$$

$$= \frac{-2}{\sqrt{2} + \sqrt{2}} = \frac{-2}{2\sqrt{2}} = \frac{-1}{\sqrt{2}}$$

Evaluate the limit if it exist

Oct 2018

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x^2 - 9x} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x^2 - 9x} * \frac{\sqrt{x} + 3}{\sqrt{x} + 3}$$

$$= \lim_{x \rightarrow 9} \frac{x - 9}{(x^2 - 9x)(\sqrt{x} + 3)}$$

$$= \lim_{x \rightarrow 9} \frac{\cancel{x - 9}}{x(\cancel{x - 9})(\sqrt{x} + 3)} = \frac{1}{9(\sqrt{9} + 3)} = \frac{1}{9(6)} = \frac{1}{54}$$

Evaluate the limit if it exist

July 2017

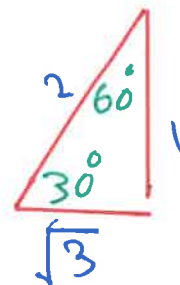
$$\lim_{x \rightarrow 1} \sin^{-1} \left(\frac{x - 1}{x^2 - 1} \right) \quad \sin^{-1} \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \sin^{-1} \left(\frac{\cancel{x - 1}}{(\cancel{x - 1})(x + 1)} \right)$$

$$= \lim_{x \rightarrow 1} \sin^{-1} \left(\frac{1}{x + 1} \right)$$

$$= \sin^{-1} \left(\frac{1}{2} \right)$$

$$= \frac{\pi}{6}$$





Text book questions



$$\lim_{x \rightarrow -1} \frac{x^2 + 2x + 1}{x^4 - 1}$$

$$\frac{1-2+1}{1-1} = \frac{0}{0}$$

$$= \lim_{x \rightarrow -1} \frac{(x+1)(x+1)}{(x^2-1)(x^2+1)}$$

$$= \lim_{x \rightarrow -1} \frac{(x+1)\cancel{(x+1)}}{(x-1)\cancel{(x+1)}(x^2+1)}$$

$$= \frac{0}{(-2)(2)} = 0$$

$$\lim_{x \rightarrow 2} \frac{x^2 - 4x + 4}{x^4 - 3x^2 - 4}$$

$$\frac{4-8+4}{16-3(4)-4} = \frac{0}{0}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)(x-2)}{(x^2+1)(x^2-4)}$$

$$= \lim_{x \rightarrow 2} \frac{(x-2)\cancel{(x-2)}}{(x^2+1)\cancel{(x-2)}(x+2)}$$

$$= \frac{0}{(4+1)(4)} = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{2x^2 - 3x + 1} \quad \frac{1-1}{2-3+1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 1} \frac{\cancel{(x-1)} (x^2 + x + 1)}{(2x - 1) \cancel{(x-1)}} = \frac{1+1+1}{2-1} = \frac{3}{1} = 3$$

Evaluate the limit if it exist

Nov 2016

$$\lim_{x \rightarrow -2} \frac{3x^2 + 5x - 2}{x^2 - 3x - 10} \quad \frac{3(4) - 10 - 2}{4 + 6 - 10} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow -2} \frac{(3x - 1) \cancel{(x + 2)}}{\cancel{(x + 2)} (x - 5)} = \frac{-6 - 1}{-2 - 5} = \frac{-7}{-7} = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow -1} \frac{x^4 - 1}{2x^2 - x - 3} \quad \frac{1-1}{2+1-3} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow -1} \frac{(x^2 - 1)(x^2 + 1)}{(2x - 3)(x + 1)}$$

$$= \lim_{x \rightarrow -1} \frac{\cancel{(x-1)} \cancel{(x+1)} (x^2 + 1)}{(2x - 3) \cancel{(x+1)}} = \frac{-2(2)}{5} = \frac{-4}{5}$$

$$1) \lim_{x \rightarrow 1} \frac{\frac{1}{x^2} - 1}{x - 1}$$

$$= \frac{\frac{1}{1} - 1}{1 - 1} = \frac{0}{0}$$

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x^2} - \frac{x^2}{x^2}}{x - 1}$$

* سبک ملحق *

$$\lim_{x \rightarrow 1} \frac{\frac{1 - x^2}{x^2}}{x - 1}$$



* سبک ملحق *

$$\lim_{x \rightarrow 1} \frac{1 - x^2}{x^2 (x - 1)}$$

$$\lim_{x \rightarrow 1} \frac{(1 - x)(1 + x)}{x^2 (x - 1)}$$

$$\lim_{x \rightarrow 1} \frac{-(x - 1)(1 + x)}{x^2 (x - 1)}$$

Now OK

$$\lim_{x \rightarrow 1} \frac{-(1 + x)}{x^2}$$

$$= \frac{-(1 + 1)}{(1)^2} = \frac{-2}{1} = -2$$



Text book questions



$$\lim_{x \rightarrow 3} \frac{\frac{1}{x} - \frac{1}{3}}{x - 3}$$

$$\lim_{x \rightarrow 3} \frac{\frac{3}{3x} - \frac{x}{3x}}{x - 3} = \lim_{x \rightarrow 3} \frac{\frac{3-x}{3x}}{x-3}$$

$$\lim_{x \rightarrow 3} \frac{3-x}{3x(x-3)}$$

$$\lim_{x \rightarrow 3} \frac{\cancel{3-x}}{3x \cancel{(x-3)}} = \frac{-1}{9}$$

$$\lim_{h \rightarrow 0} \frac{(3+h)^{-1} - 3^{-1}}{h}$$

$$\lim_{h \rightarrow 0} \frac{\frac{1}{3+h} - \frac{1}{3}}{h} \quad \frac{0}{0}$$

$$\lim_{h \rightarrow 0} \frac{\frac{3}{3(3+h)} - \frac{3+h}{3(3+h)}}{h} = \lim_{h \rightarrow 0} \frac{\frac{3-(3+h)}{3(3+h)}}{h}$$

$$\lim_{h \rightarrow 0} \frac{3-3-h}{3(3+h)h}$$

$$\lim_{h \rightarrow 0} \frac{\cancel{-h}}{3(3+h)\cancel{h}} = \frac{-1}{3(3)} = -\frac{1}{9}$$



Text book questions



$$\lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h}$$

0/0

$$\lim_{h \rightarrow 0} \frac{\frac{x^2}{x^2(x+h)^2} - \frac{(x+h)^2}{x^2(x+h)^2}}{h} = \lim_{h \rightarrow 0} \frac{\frac{x^2 - (x+h)^2}{x^2(x+h)^2}}{h}$$

$$\lim_{h \rightarrow 0} \frac{x^2 - (x+h)^2}{x^2(x+h)^2 \cdot h}$$

$$\lim_{h \rightarrow 0} \frac{[x - (x+h)][x + (x+h)]}{x^2(x+h)^2 h} = \lim_{h \rightarrow 0} \frac{[x - x - h][2x+h]}{x^2(x+h)^2 h}$$

$$\lim_{h \rightarrow 0} \frac{-h[2x+h]}{x^2(x+h)^2 h} = \frac{-[2x+0]}{x^2(x+0)^2} = \frac{-2x}{x^4} = \frac{-2}{x^3}$$

$$\lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

0/0

$$\lim_{h \rightarrow 0} \frac{[(x+h) - x][(x+h)^2 + (x+h)x + x^2]}{h}$$

$$\lim_{h \rightarrow 0} \frac{h[(x+h)^2 + (x+h)x + x^2]}{h}$$

$$= (x+0)^2 + (x+0)x + x^2 = x^2 + x^2 + x^2 = 3x^2$$

$$\begin{array}{l} x-1 \\ 1-x \end{array}$$

$$9) \lim_{x \rightarrow 1} \left[\frac{1}{x-1} - \frac{2}{x^2-1} \right]$$

$$= \frac{1}{0} - \frac{2}{0}$$

$$= \infty - \infty$$

$$\lim_{x \rightarrow 1} \left[\frac{1}{x-1} - \frac{2}{(x-1)(x+1)} \right]$$

$$\lim_{x \rightarrow 1} \left[\frac{(x+1)}{(x-1)(x+1)} - \frac{2}{(x-1)(x+1)} \right]$$

$$\lim_{x \rightarrow 1} \left[\frac{x+1-2}{(x-1)(x+1)} \right]$$

$$\lim_{x \rightarrow 1} \left[\frac{\cancel{x-1}}{\cancel{(x-1)}(x+1)} \right]$$

$$\lim_{x \rightarrow 1} \left[\frac{1}{x+1} \right]$$

$$= \frac{1}{1+1} = \frac{1}{2}$$

$$\lim_{x \rightarrow 4} \left[\frac{1}{x-4} - \frac{3}{x^2 - 5x + 4} \right]$$

$$\frac{1}{0} - \frac{1}{0}$$

$$\infty - \infty$$

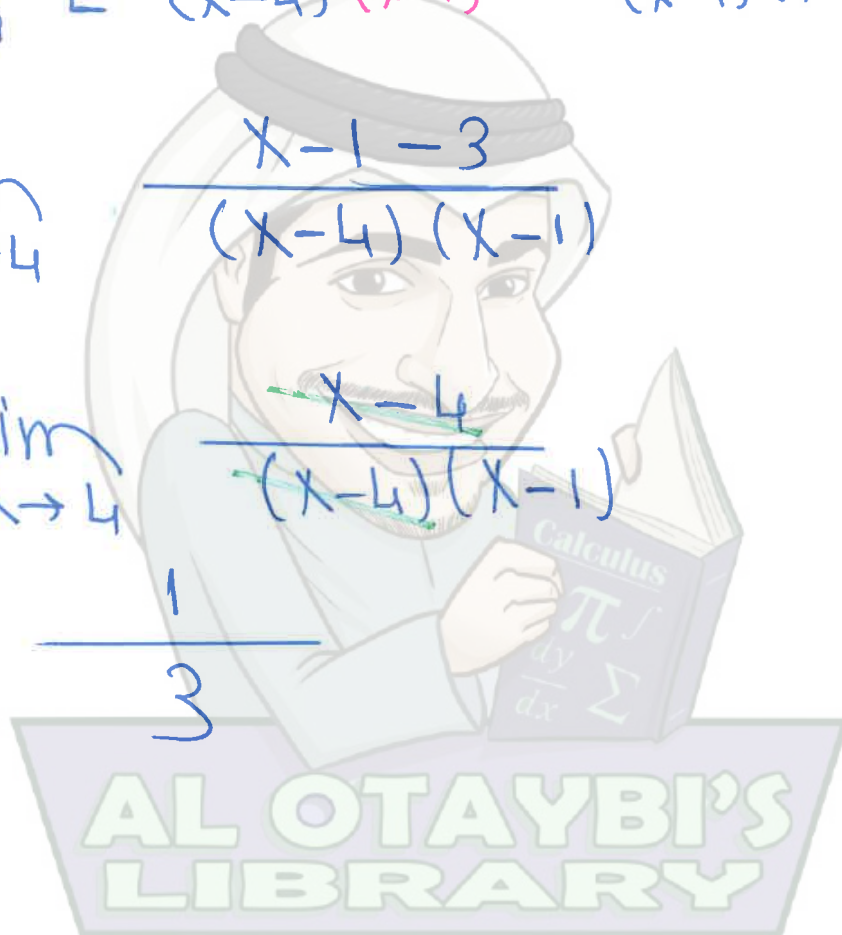
$$= \lim_{x \rightarrow 4} \left[\frac{1}{x-4} - \frac{3}{(x-1)(x-4)} \right]$$

$$= \lim_{x \rightarrow 4} \left[\frac{x-1}{(x-4)(x-1)} - \frac{3}{(x-1)(x-4)} \right]$$

$$= \lim_{x \rightarrow 4} \frac{x-1-3}{(x-4)(x-1)}$$

$$= \lim_{x \rightarrow 4} \frac{\cancel{x-4}}{\cancel{(x-4)}(x-1)}$$

$$= \frac{1}{3}$$



Evaluate the limit if it exist

Apr 2017

$$\lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{4}{x^2-4} \right) \quad \frac{1}{0} - \frac{4}{0}$$

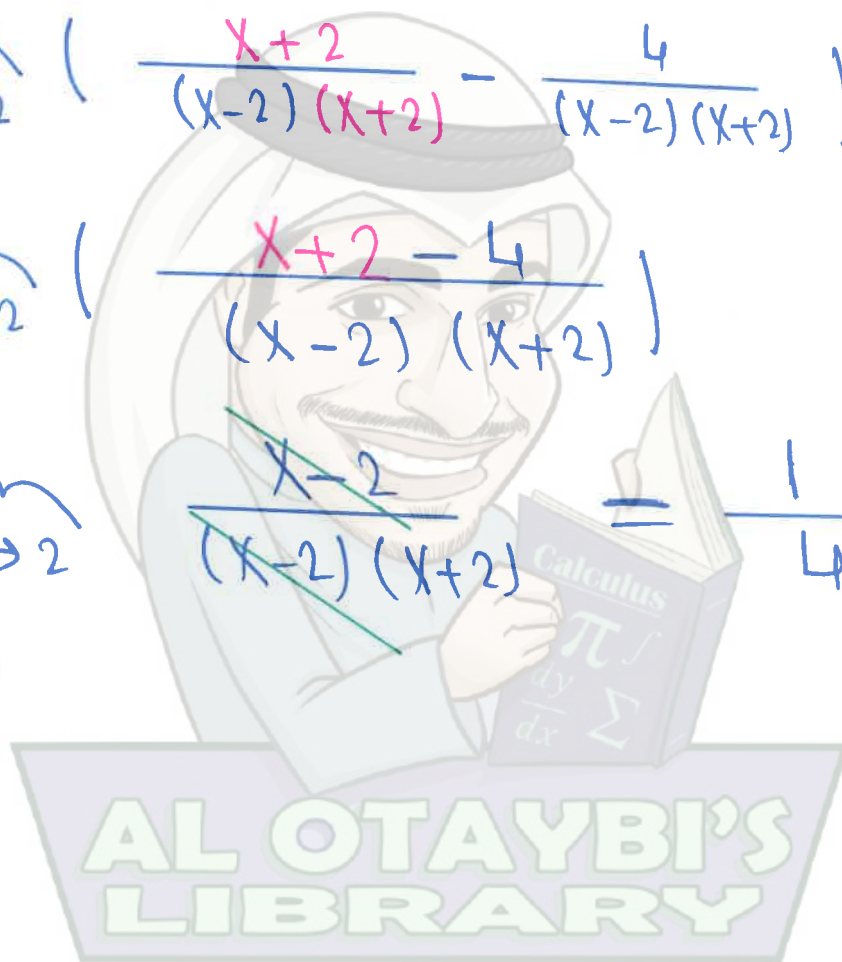
$\infty - \infty$

$$= \lim_{x \rightarrow 2} \left(\frac{1}{x-2} - \frac{4}{(x-2)(x+2)} \right)$$

$$= \lim_{x \rightarrow 2} \left(\frac{x+2}{(x-2)(x+2)} - \frac{4}{(x-2)(x+2)} \right)$$

$$= \lim_{x \rightarrow 2} \left(\frac{x+2-4}{(x-2)(x+2)} \right)$$

$$= \lim_{x \rightarrow 2} \frac{\cancel{x-2}}{(\cancel{x-2})(x+2)} = \frac{1}{4}$$





Text book questions



$$\lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{1}{t} \right)$$

$$\frac{1}{0} - \frac{1}{0}$$
$$\infty - \infty$$

$$= \lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{\sqrt{1+t}}{t\sqrt{1+t}} \right)$$

$$= \lim_{t \rightarrow 0} \frac{1 - \sqrt{1+t}}{t\sqrt{1+t}} \quad \frac{0}{0}$$

$$= \lim_{t \rightarrow 0} \frac{1 - \sqrt{1+t}}{t\sqrt{1+t}} * \frac{1 + \sqrt{1+t}}{1 + \sqrt{1+t}}$$

$$= \lim_{t \rightarrow 0} \frac{1 - (1+t)}{t\sqrt{1+t} [1 + \sqrt{1+t}]}$$

$$= \lim_{t \rightarrow 0} \frac{1 - 1 - t}{t\sqrt{1+t} [1 + \sqrt{1+t}]}$$

$$= \lim_{t \rightarrow 0} \frac{-t}{t\sqrt{1+t} [1 + \sqrt{1+t}]}$$

$$= \frac{-1}{\sqrt{1} [1 + \sqrt{1}]} = \frac{-1}{2}$$



Nice questions !



Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{e^x - 1}$$

$$\frac{1-1}{1-1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{(e^x - 1)}(e^x + 1)}{\cancel{e^x - 1}} = e^0 + 1 = 1 + 1 = 2$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{5^{3x} - 1}{5^x - 1}$$

$$\frac{1-1}{1-1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{(5^x - 1)}((5^x)^2 + 5^x + 1)}{\cancel{5^x - 1}} = (5^0)^2 + 5^0 + 1 = 1 + 1 + 1 = 3$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{2^x - 1}{4^x - 1}$$

$$\frac{1-1}{1-1} \quad \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{2^x - 1}{(2^2)^x - 1}$$

$$= \lim_{x \rightarrow 0} \frac{2^x - 1}{2^{2x} - 1} = \lim_{x \rightarrow 0} \frac{\cancel{2^x - 1}}{\cancel{(2^x - 1)}(2^x + 1)} = \frac{1}{1+1} = \frac{1}{2}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{8^x - 1}{2^x - 1} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{2^{3x} - 1}{2^x - 1}$$

$$\lim_{x \rightarrow 0} \frac{(2^x - 1)(2^x)^2 + 2^x + 1}{2^x - 1}$$

$$= 1 + 1 + 1 = 3$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{e^{2x} - e^x}{e^x - 1} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{e^x(e^x - 1)}{e^x - 1}$$

$$= e^0 = 1$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{1 - 3^x}{27^x - 1} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{1 - 3^x}{3^{3x} - 1}$$

$$\lim_{x \rightarrow 0} \frac{1 - 3^x}{(3^x - 1)(3^x)^2 + 3^x + 1}$$

$$\lim_{x \rightarrow 0} \frac{-(3^x - 1)}{(3^x - 1)(3^x)^2 + 3^x + 1} = \frac{-1}{1 + 1 + 1} = -\frac{1}{3}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} \frac{e^{2x} + e^x - 2}{e^x - 1} \quad \frac{0}{0}$$

$$\lim_{x \rightarrow 0} \frac{(e^x - 1)(e^x + 2)}{e^x - 1}$$

$$= e^0 + 2 = 1 + 2 = 3$$



Text book questions



Given that

$$\lim_{x \rightarrow 2} f(x) = 4, \quad \lim_{x \rightarrow 2} g(x) = -2, \quad \lim_{x \rightarrow 2} h(x) = 0$$

Find the limits that exist. If the limit does not exist, explain why.

a) $\lim_{x \rightarrow 2} [f(x) + 5g(x)]$ b) $\lim_{x \rightarrow 2} [g(x)]^3$

c) $\lim_{x \rightarrow 2} \sqrt{f(x)}$ d) $\lim_{x \rightarrow 2} \frac{3f(x)}{g(x)}$

e) $\lim_{x \rightarrow 2} \frac{g(x)}{h(x)}$ f) $\lim_{x \rightarrow 2} \frac{g(x)h(x)}{f(x)}$

a) $\lim_{x \rightarrow 2} [f(x) + 5g(x)] = 4 + 5(-2) = 4 - 10 = -6$

b) $\lim_{x \rightarrow 2} [g(x)]^3 = (-2)^3 = -8$

c) $\lim_{x \rightarrow 2} \sqrt{f(x)} = \sqrt{4} = 2$

d) $\lim_{x \rightarrow 2} \frac{3f(x)}{g(x)} = \frac{3(4)}{-2} = -6$

e) $\lim_{x \rightarrow 2} \frac{g(x)}{h(x)} = \frac{-2}{0} = -\infty$ (DNE)

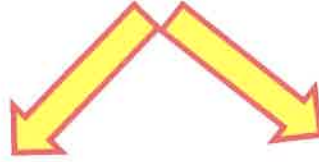
f) $\lim_{x \rightarrow 2} \frac{g(x)h(x)}{f(x)} = \frac{-2(0)}{4} = 0$



Sandwich Theorem (Squeeze Theorem)



نظرية تستخدم لحل مسائل اللمت اذا كان



طلب اللمت لدالة مجهولة مع شرط
وجود المتباينة
 $.... < f(x) <$

ناتج التعويض المباشر يحتوي على
 $\sin \infty$ or $\cos \infty$

اذا كان الأس زوجي

$$0 \leq \sin^n \theta \leq 1$$

$$0 \leq \cos^n \theta \leq 1$$

اذا كان الأس فردي

$$-1 \leq \sin^n \theta \leq 1$$

$$-1 \leq \cos^n \theta \leq 1$$



متى يلزم تغيير اتجاه المتباينة ؟؟



عند قلب البسط والمقام

$$3 \leq x \leq 7$$

$$\Rightarrow \frac{1}{3} \geq \frac{1}{x} \geq \frac{1}{7}$$

عند الضرب أو القسمة في عدد سالب

$$-1 \leq -x \leq 7$$

$$\Rightarrow \begin{matrix} * -1 \\ 1 \geq x \geq -7 \end{matrix}$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$$

$$0 \sin \frac{1}{0}$$
$$0 \sin \infty$$

Sandwich

$$-1 \leq \sin \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

$$\therefore x^2 \geq 0$$

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

$$\lim_{x \rightarrow 0} -x^2 = 0$$

$$\lim_{x \rightarrow 0} x^2 = 0$$

$\therefore$ By Sandwich Theorem

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$$\lim_{x \rightarrow 0} x^2 \sin^2 \left(\frac{1}{x} \right)$$

$$\begin{aligned} &0 \sin^2 \frac{1}{0} \\ &0 \sin^2 \infty \end{aligned}$$

$$0 \leq \sin^2 \theta \leq 1$$

$$0 \leq \sin^2 \left(\frac{1}{x} \right) \leq 1$$

$$\because x^2 \geq 0$$

$$0 \leq x^2 \sin^2 \left(\frac{1}{x} \right) \leq x^2$$

$$\lim_{x \rightarrow 0} 0 = 0$$

$$\lim_{x \rightarrow 0} x^2 = 0$$

$$\therefore \lim_{x \rightarrow 0} 0 = \lim_{x \rightarrow 0} x^2 = 0$$

By Sandwich Theorem

$$\lim_{x \rightarrow 0} x^2 \sin^2 \left(\frac{1}{x} \right) = 0$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 3} \left[5 + |x - 3| \sin \left(\frac{\pi}{x - 3} \right) \right]$$

$$5 + 0 \sin \frac{\pi}{0}$$

$$-1 \leq \sin \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \sin \left(\frac{\pi}{x-3} \right) \leq 1$$

$$\because |x-3| > 0$$

$$-|x-3| \leq |x-3| \sin \left(\frac{\pi}{x-3} \right) \leq |x-3|$$

$$\lim_{x \rightarrow 3} -|x-3| = 0$$

$$\lim_{x \rightarrow 3} |x-3| = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{x \rightarrow 3} |x-3| \sin \left(\frac{\pi}{x-3} \right) = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow 3} \left[5 + |x-3| \sin \left(\frac{\pi}{x-3} \right) \right] = 5 + 0 = 5$$

Evaluate the limit if it exist

$$\lim_{x \rightarrow 1} \left[9x - (x - 1)^2 \sin \left(\frac{1}{1-x} \right) \right]$$

$$9 - 0 \sin \frac{1}{0}$$

$$-1 \leq \sin \left(\frac{1}{1-x} \right) \leq 1$$

$$\because (x-1)^2 > 0$$

$$-(x-1)^2 \leq (x-1)^2 \sin \left(\frac{1}{1-x} \right) \leq (x-1)^2$$

$$\lim_{x \rightarrow 1} -(x-1)^2 = 0$$

$$\lim_{x \rightarrow 1} (x-1)^2 = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{x \rightarrow 1} (x-1)^2 \sin \left(\frac{1}{1-x} \right) = 0$$

$$\Rightarrow \therefore \lim_{x \rightarrow 1} \left[9x - (x-1)^2 \sin \left(\frac{1}{1-x} \right) \right] = 9 - 0 = 9$$

$$\lim_{x \rightarrow 0} \left(\sqrt{3+x^2} + \frac{x^2}{\csc\left(\frac{1}{x}\right)} \right)$$

$$\lim_{x \rightarrow 0} \left(\sqrt{3+x^2} + x^2 \sin\left(\frac{1}{x}\right) \right)$$

$$\sqrt{3+0}$$

$$\begin{matrix} 0 \sin \frac{1}{0} \\ 0 \sin \infty \end{matrix}$$

$$-1 \leq \sin \theta \leq 1$$

$$-1 \leq \sin \frac{1}{x} \leq 1$$

$$\because x^2 > 0$$

$$-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

$$\lim_{x \rightarrow 0} -x^2 = 0$$

$$\lim_{x \rightarrow 0} x^2 = 0$$

$$\therefore \lim_{x \rightarrow 0} -x^2 = \lim_{x \rightarrow 0} x^2 = 0$$

By Sandwich Theorem

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} \left(\sqrt{3+x^2} + x^2 \sin \frac{1}{x} \right) = \sqrt{3} + 0 = \sqrt{3}$$

Evaluate the limit if it exist

Oct 2018

$$\lim_{x \rightarrow 1} |x - 1| \sin\left(\frac{1}{x^2 - 1}\right)$$

$$\begin{aligned} 0 \sin \frac{1}{0} \\ 0 \sin \infty \end{aligned}$$

$$-1 \leq \sin \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \sin \frac{1}{x^2 - 1} \leq 1$$

$$\therefore |x - 1| \geq 0$$

$$-|x - 1| \leq |x - 1| \sin \frac{1}{x^2 - 1} \leq |x - 1|$$

$$\lim_{x \rightarrow 1} -|x - 1| = 0$$

$$\lim_{x \rightarrow 1} |x - 1| = 0$$

By Sandwich Theorem

$$\lim_{x \rightarrow 1} |x - 1| \sin\left(\frac{1}{x^2 - 1}\right) = 0$$

Evaluate the limit if it exist

Apr 2017

$$\lim_{x \rightarrow 0} (x - 2)^2 \sin\left(\frac{\pi}{x - 2}\right)$$

$$\begin{aligned} 0 \sin\left(\frac{\pi}{0}\right) \\ 0 \sin \infty \end{aligned}$$

$$-1 \leq \sin\left(\frac{\pi}{x - 2}\right) \leq 1$$

!!

$$\begin{aligned} &= (0 - 2)^2 \sin\left(\frac{\pi}{-2}\right) \\ &= 4 \sin\left(-\frac{\pi}{2}\right) \\ &= -4 \sin \frac{\pi}{2} \\ &= -4 (1) = -4 \end{aligned}$$

$$\begin{aligned} &+ (0, 1) \\ &+ \end{aligned}$$



Text book questions



Prove that $\lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} = 0$

$$\lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} \quad \begin{matrix} 0 \cos \frac{2}{0} \\ 0 \cos \infty \end{matrix}$$

$$-1 \leq \cos \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \cos \frac{2}{x} \leq 1$$

$$\therefore x^4 \geq 0$$

$$-x^4 \leq x^4 \cos \frac{2}{x} \leq x^4$$

$$\lim_{x \rightarrow 0} -x^4 = 0$$

$$\lim_{x \rightarrow 0} x^4 = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{x \rightarrow 0} x^4 \cos \frac{2}{x} = 0$$

Prove that $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\frac{\pi}{x})} = 0$

$$\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\frac{\pi}{x})} \quad \begin{matrix} 0 e^{\sin \frac{\pi}{0}} \\ 0 e^{\sin \infty} \end{matrix}$$

$$-1 \leq \sin \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \sin \frac{\pi}{x} \leq 1$$

$$e^{-1} \leq e^{\sin \frac{\pi}{x}} \leq e^1$$

$$\therefore \sqrt{x} \geq 0$$

$$\sqrt{x} e^{-1} \leq \sqrt{x} e^{\sin \frac{\pi}{x}} \leq \sqrt{x} e^1$$

$$\lim_{x \rightarrow 0} \sqrt{x} e^{-1} = 0$$

$$\lim_{x \rightarrow 0} \sqrt{x} e^1 = 0$$

$\therefore$ By Sandwich Theorem $\rightarrow \lim_{x \rightarrow 0} \sqrt{x} e^{\sin \frac{\pi}{x}} = 0$

Evaluate the limit if it exist

$$\lim_{\theta \rightarrow 0} \frac{\theta^2}{2 + \cos\left(\frac{1}{\theta}\right)}$$

$$\frac{0}{2 + \cos \frac{1}{0}}$$
$$\frac{0}{2 + \cos \infty}$$

$$-1 \leq \cos \theta \leq 1 \quad \forall \theta$$

$$-1 \leq \cos\left(\frac{1}{\theta}\right) \leq 1$$

+2

$$1 \leq 2 + \cos\left(\frac{1}{\theta}\right) \leq 3$$

$$\frac{1}{1} \geq \frac{\theta^2}{2 + \cos\left(\frac{1}{\theta}\right)} \geq \frac{\theta^2}{3}$$

$$\because \theta^2 \geq 0$$

$$\theta^2 \geq \frac{\theta^2}{2 + \cos\left(\frac{1}{\theta}\right)} \geq \frac{\theta^2}{3}$$

$$\lim_{\theta \rightarrow 0} \theta^2 = 0$$

$$\lim_{\theta \rightarrow 0} \frac{\theta^2}{3} = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{\theta \rightarrow 0} \frac{\theta^2}{2 + \cos\left(\frac{1}{\theta}\right)} = 0$$



Text book questions



If $4x - 9 \leq f(x) \leq x^2 - 4x + 7$ for $x \geq 0$.

Find $\lim_{x \rightarrow 4} f(x)$

$$4x - 9 \leq f(x) \leq x^2 - 4x + 7$$

$$\lim_{x \rightarrow 4} (4x - 9) = 16 - 9 = 7$$

$$\lim_{x \rightarrow 4} (x^2 - 4x + 7) = 16 - 16 + 7 = 7$$

∴ By Sandwich Theorem

$$\lim_{x \rightarrow 4} f(x) = 7$$

If $2x \leq g(x) \leq x^4 - x^2 + 2$ for all x

Evaluate $\lim_{x \rightarrow 1} g(x)$

$$2x \leq g(x) \leq x^4 - x^2 + 2$$

$$\lim_{x \rightarrow 1} 2x = 2$$

$$\lim_{x \rightarrow 1} (x^4 - x^2 + 2) = (1)^4 - (1)^2 + 2 = 2$$

∴ By Sandwich Theorem

$$\lim_{x \rightarrow 1} g(x) = 2$$

Assume $\lim_{x \rightarrow -1} f(x)$ exist and

$$\frac{x^2 + x - 2}{x + 3} \leq \frac{f(x)}{(x - 1)^2} \leq \frac{x^2 + 2x - 1}{x + 3}$$

Find $\lim_{x \rightarrow -1} \frac{f(x)}{(x - 1)^2}$

$$\lim_{x \rightarrow -1} \frac{x^2 + x - 2}{x + 3}$$

$$= \frac{1 - 1 - 2}{-1 + 3} = \frac{-2}{2} = -1$$

$$\lim_{x \rightarrow -1} \frac{x^2 + 2x - 1}{x + 3}$$

$$= \frac{1 - 2 - 1}{-1 + 3} = \frac{-2}{2} = -1$$

$\Rightarrow \therefore$ By Sandwich Theorem

$$\lim_{x \rightarrow -1} \frac{f(x)}{(x - 1)^2} = -1$$

Assume $\lim_{x \rightarrow -1} f(x)$ exist and

$$\frac{x^2 + x - 2}{x + 3} \leq \frac{f(x)}{(x - 1)^2} \leq \frac{x^2 + 2x - 1}{x + 3}$$

Find $\lim_{x \rightarrow -1} f(x)$

$$\therefore (x - 1)^2 > 0$$

$$(x - 1)^2 \frac{x^2 + x - 2}{x + 3} \leq \cancel{(x - 1)^2} \frac{f(x)}{\cancel{(x - 1)^2}} \leq (x - 1)^2 \frac{x^2 + 2x - 1}{x + 3}$$

$$\lim_{x \rightarrow -1} (x - 1)^2 \frac{x^2 + x - 2}{x + 3}$$

$$= (-2)^2 \frac{1 - 1 - 2}{-1 + 3} = 4 \frac{-2}{2} = -4$$

$$\lim_{x \rightarrow -1} (x - 1)^2 \frac{x^2 + 2x - 1}{x + 3}$$

$$= (-2)^2 \frac{1 - 2 - 1}{-1 + 3} = 4 \frac{-2}{2} = -4$$

$\Rightarrow \therefore$ By Sandwich Theorem $\lim_{x \rightarrow -1} f(x) = -4$

If $e^x \leq f(x) \leq \frac{4x^2 + 1}{x^2}$, find $\lim_{x \rightarrow 0} x^4 f(x)$, if it exist

Mar 2019

$$e^x \leq f(x) \leq \frac{4x^2 + 1}{x^2}$$

$$\therefore x^4 > 0$$

$$x^4 e^x \leq x^4 f(x) \leq x^4 \frac{4x^2 + 1}{x^2}$$

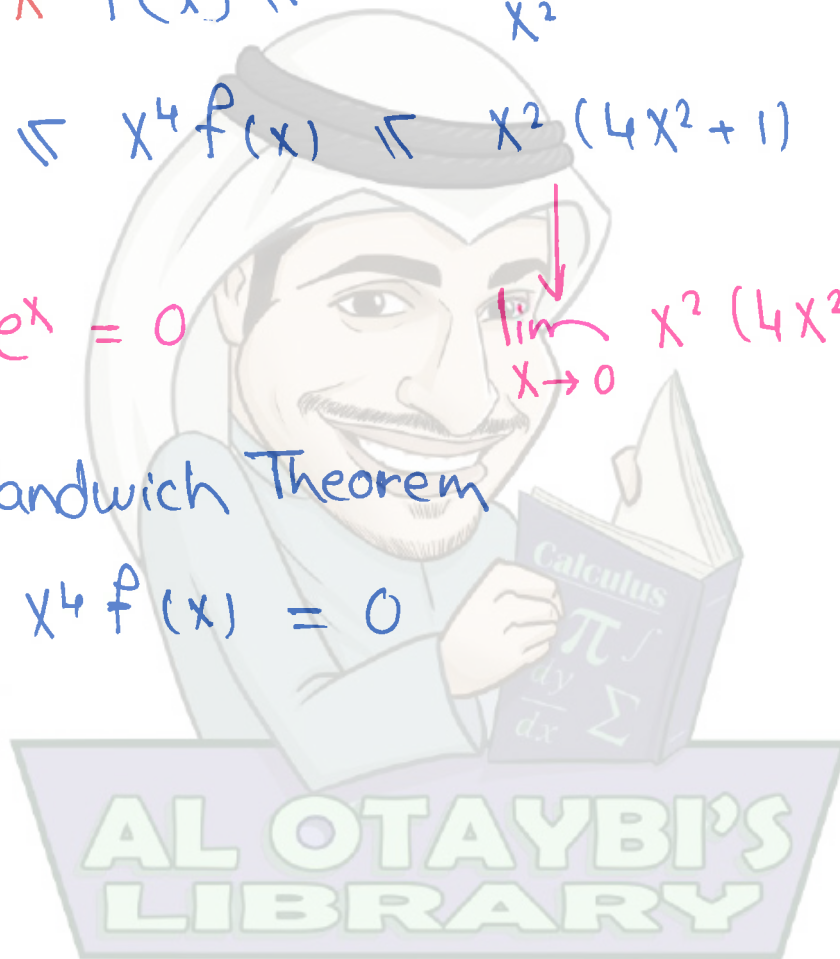
$$x^4 e^x \leq x^4 f(x) \leq x^2 (4x^2 + 1)$$

$$\downarrow$$
$$\lim_{x \rightarrow 0} x^4 e^x = 0$$

$$\downarrow$$
$$\lim_{x \rightarrow 0} x^2 (4x^2 + 1) = 0$$

$\therefore$ By Sandwich Theorem

$$\lim_{x \rightarrow 0} x^4 f(x) = 0$$



be a function satisfying

$$4x - 9 \leq f(x) + x \leq x^2 - 4x + 7$$

for all $x \in (-\infty, \infty)$

Find $\lim_{x \rightarrow 4} f(x)$ if it exist

$$4x - 9 \leq f(x) + x \leq x^2 - 4x + 7$$

$$4x - 9 - x \leq f(x) \leq x^2 - 4x + 7 - x$$

$$3x - 9 \leq f(x) \leq x^2 - 5x + 7$$

$$\lim_{x \rightarrow 4} (3x - 9) = 12 - 9 = 3$$

$$\lim_{x \rightarrow 4} (x^2 - 5x + 7) = 16 - 20 + 7 = 3$$

$$\therefore \lim_{x \rightarrow 4} (3x - 9) = \lim_{x \rightarrow 4} (x^2 - 5x + 7) = 3$$

$\Rightarrow$ By Sandwich Theorem
$$\lim_{x \rightarrow 4} f(x) = 3$$

Assume that $\lim_{x \rightarrow 1} \frac{1}{f(x)}$ exist and

$$\frac{x^2 + x - 1}{x + 2} \leq \frac{(x + 1)^2}{f(x)} \leq \frac{x^2 - 2x + 2}{x + 2}$$

Find $\lim_{x \rightarrow 1} f(x)$

$$(x+1)^2 > 0$$

$$\div (x+1)^2$$

$$\frac{1}{(x+1)^2} \cdot \frac{x^2 + x - 1}{x + 2} \leq \frac{1}{f(x)} \leq \frac{1}{(x+1)^2} \cdot \frac{x^2 - 2x + 2}{x + 2}$$

$$\frac{(x+1)^2}{1} \cdot \frac{x+2}{x^2 + x - 1} \geq f(x) \geq \frac{(x+1)^2}{1} \cdot \frac{x+2}{x^2 - 2x + 2}$$

$$\lim_{x \rightarrow 1} (x+1)^2 \cdot \frac{x+2}{x^2 + x - 1} = (2)^2 \cdot \frac{3}{1+1-1} = 4 \left(\frac{3}{1} \right) = 12$$

$$\lim_{x \rightarrow 1} (x+1)^2 \cdot \frac{x+2}{x^2 - 2x + 2} = (2)^2 \cdot \frac{3}{1-2+2} = 4(3) = 12$$

$$\therefore \lim_{x \rightarrow 1} (x+1)^2 \cdot \frac{x+2}{x^2 + x - 1} = \lim_{x \rightarrow 1} (x+1)^2 \cdot \frac{x+2}{x^2 - 2x + 2} = 12$$

$\Rightarrow$ By Sandwich Theorem

$$\lim_{x \rightarrow 1} f(x) = 12$$



Text book questions



If $\lim_{x \rightarrow 1} \frac{f(x) - 8}{x - 1} = 10$. Find $\lim_{x \rightarrow 0} f(x)$

$$f(x) - 8 = \frac{f(x) - 8}{x - 1} (x - 1)$$

$$\lim_{x \rightarrow 1} [f(x) - 8] = \lim_{x \rightarrow 1} \left[\frac{f(x) - 8}{x - 1} (x - 1) \right]$$

$$\lim_{x \rightarrow 1} [f(x) - 8] = 10 (0)$$

$$\lim_{x \rightarrow 1} f(x) - \lim_{x \rightarrow 1} 8 = 0$$

$$\Rightarrow \lim_{x \rightarrow 1} f(x) = 8$$

If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 5$. find the following limits

a) $\lim_{x \rightarrow 0} f(x)$

b) $\lim_{x \rightarrow 0} \frac{f(x)}{x}$

$$(a) f(x) = \frac{f(x)}{x^2} \cdot x^2 \quad (b) \frac{f(x)}{x} = \frac{f(x)}{x \cdot x} \cdot x$$

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left[\frac{f(x)}{x^2} \cdot x^2 \right]$$

$$\lim_{x \rightarrow 0} f(x) = 5(0)$$

$$\Rightarrow \lim_{x \rightarrow 0} f(x) = 0$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \left[\frac{f(x)}{x^2} \cdot x \right]$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = 5(0) = 0$$

$$\lim_{x \rightarrow 1} (x-1)^9 \cos\left(\frac{1}{x-1}\right)$$

$$\begin{aligned} 0 &\leq \cos \frac{1}{0} \\ 0 &\leq \cos \infty \end{aligned}$$

$$-1 \leq \cos \theta \leq 1$$

$$-1 \leq \cos\left(\frac{1}{x-1}\right) \leq 1$$

$$\text{as } x \longrightarrow 1^+$$

$$\Rightarrow (x-1)^9 \rightarrow 0$$

$$-(x-1)^9 \leq (x-1)^9 \cos\left(\frac{1}{x-1}\right) \leq (x-1)^9$$

$$\lim_{x \rightarrow 1^+} -(x-1)^9 = 0$$

$$\lim_{x \rightarrow 1^+} (x-1)^9 = 0$$

∴ By Sandwich Theorem

$$\lim_{x \rightarrow 1^+} f(x) = 0$$

$$\text{as } x \longrightarrow 1^-$$

$$\Rightarrow (x-1)^9 \leq 0$$

$$-(x-1)^9 \leq (x-1)^9 \cos\left(\frac{1}{x-1}\right) \leq (x-1)^9$$

$$\lim_{x \rightarrow 1^-} -(x-1)^9 = 0$$

$$\lim_{x \rightarrow 1^-} (x-1)^9 = 0$$

∴ By Sandwich Theorem

$$\lim_{x \rightarrow 1^-} f(x) = 0$$

$$\therefore \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x) = 0$$

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$$\Rightarrow \therefore \lim_{x \rightarrow 1} f(x) = 0$$